SALZMAN BASEBALL

GPP & SPP MASTER BLUEPRINT

Movement-Pattern Programming • Loads • Sets • Reps • Rest Periods

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Version 2 — Revised Weekly Distribution & Training Prescriptions

This revision adds a Saturday training day to distribute weekly workload, replaces RPE-only prescriptions with target loading ranges where measurable, separates isometric methods by intent and duration, updates complex-training recovery, adds year-round power exposures, and expands forearm development beyond pinch work.

Programming language: Exercises are described by their primary movement or physical-capacity category (for example, *hinge pattern*, *horizontal push pattern*, or *finger flexor capacity*) rather than by a specific exercise name. Exercise selection remains flexible within each category.

Loading note: Percentages refer to a reliable, current 1RM only when the movement has a repeatable and appropriate 1RM test. For unilateral, rotational, isometric, and ballistic categories without a valid 1RM, select a load that permits the prescribed output, range of motion, and technique. Do not force a percentage onto a non-standardized movement.

1. GENERAL PREPARATORY PHASE (GPP — EARLY OFF-SEASON)

Primary objective: Build foundational structural capacity, restore/maintain useful joint motion, improve local tissue tolerance, and establish a base of aerobic and movement capacity while introducing low-volume power work from the beginning.

Phase parameters: Five training days distributed across Monday, Tuesday, Thursday, and Saturday, with Wednesday and Friday reserved for recovery. Most strength work is submaximal. Power work is low volume and high quality.

Monday — Lower-Body Structural Capacity & Unilateral Hip Function

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Autonomic / thoracic preparation	Low intensity; nasal or controlled breathing	8–12 min	As needed	Breathing, thoracic mobility, foot/ankle sensory preparation; should improve readiness, not create fatigue.
Unilateral hip internal-rotation pattern	Light–moderate constant-tension load; controlled	3 × 8–10/side	45–60 sec	3-sec eccentric; 1–2-sec end-range pause.
Split-stance yielding isometric pattern	Moderate load; position held without shaking or loss of alignment	3 × 20–30 sec/leg	60–90 sec	Use a tolerable lengthened position; stop before form deteriorates.
Hinge pattern — structural strength	60–70% 1RM where applicable	3–4 × 8	2 min	Controlled 2–3-sec lowering; leave several quality reps in reserve.
Posterior-chain long-length isometric	Bodyweight to light external load	2–3 × 30–45 sec	60–90 sec	Submaximal positional hold; not a maximal effort.
Zone 2 aerobic capacity	Conversational pace	20–30 min	—	Continuous low-intensity modality; maintain conversational breathing.

Tuesday — Upper-Body Structure & Forearm Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Rotational external-rotation pattern	Light–moderate constant-tension load	3 × 10–12/arm	60 sec	Controlled 3–4-sec eccentric; no momentum.
External-rotation yielding isometric	Light constant-tension load	3 × 20–25 sec/arm	60 sec	Steady tension; maintain joint position.
Unilateral horizontal push pattern	65–75% 1RM where applicable	3–4 × 8–10/arm	90–120 sec	Controlled lowering; stable trunk and shoulder blade.
Vertical pull pattern	65–75% 1RM where applicable	3–4 × 8–10	90–120 sec	Full controlled range; avoid grinding repetitions.
Wrist flexion capacity	Moderate load; approximately 65–75% of standardized exercise max	3 × 8–12	60–90 sec	Controlled full range; progressive loading.
Wrist extension capacity	Moderate load; approximately 65–75% of standardized exercise max	3 × 10–12	60–90 sec	Controlled eccentric; balance flexor work.
Finger extension capacity	Light–moderate resistance	2–3 × 12–20	45–60 sec	Full opening range; smooth repetitions.

Wednesday — Recovery & Movement Quality

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Zone 2 aerobic recovery	Easy conversational pace	20–30 min	—	Low fatigue; should leave the athlete feeling better.
Mobility / breathing / balance	Low intensity	10–15 min	As needed	Use individual restrictions and movement-quality needs; no forced end-range work.
Soft-quality movement screens	Submaximal	1–2 rounds	As needed	Single-leg balance, active/passive rotational range comparison, and relaxed breathing checks.

Thursday — Deceleration, Adductor Capacity & Rotational Control

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Lateral bound / frontal-plane power pattern	Bodyweight; maximal intent with clean landing	3 × 3/side	90–120 sec	Power exposure; reset fully between repetitions.
Rotational ballistic throw pattern	Light implement selected for speed	3 × 3/side	90–120 sec	Fast intent; terminate if speed or coordination clearly declines.
Adductor yielding isometric	Bodyweight to moderate external load	3 × 20–30 sec/side	60–90 sec	Maintain pelvis and trunk position.
Eccentric deceleration / drop-to-stick pattern	Bodyweight or light external load	3 × 6/leg	90 sec	3-sec eccentric; controlled landing and stable knee/hip alignment.
Upright squat pattern	60–70% 1RM where applicable	3–4 × 8	2 min	Controlled tempo; choose a non-back-squat variation consistent with program constraints.
Rotational eccentric capacity	Light–moderate constant-tension load	2–3 × 8–10/side	60–90 sec	3–4-sec eccentric; smooth controlled return.

Friday — Recovery

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Zone 1–2 aerobic flush (optional)	Very easy	15–25 min	—	Use only when it improves recovery; otherwise rest.
Mobility / breathing	Low intensity	10–15 min	As needed	No strenuous conditioning or high-intent work.

Saturday — Scapular Function, Upper-Body Structure & Forearm Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Horizontal pull pattern	65–75% 1RM where applicable	3–4 × 8–10	90–120 sec	Constant-tension loading; controlled full range.
Scapular upward-rotation pattern	Light–moderate load	3 × 10–12	60 sec	Emphasize smooth upward rotation and control.
Unilateral horizontal push pattern	65–70% 1RM where applicable	3 × 8–10/arm	90 sec	Moderate volume; avoid failure.
Ulnar-deviation capacity	Moderate load	3 × 8–12/side	60–90 sec	Controlled range; avoid using momentum.
Finger-flexion capacity	Moderate load	3 × 8–12	60–90 sec	Train finger flexors through a repeatable range.
Explosive pinch closure	Load that permits fast closure	3 × 5–8	60–90 sec	Maximal speed; stop if closure speed falls.
Two-point pinch hold	Moderate load; submaximal	2 × 10–15 sec	60 sec	Thumb-to-index emphasis; clean hold.

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Three-point pinch hold	Moderate load; submaximal	2 × 10–15 sec	60 sec	Thumb-to-index/middle finger emphasis; clean hold.

2. SPECIFIC PREPARATORY PHASE (SPP — LATE OFF-SEASON)

Primary objective: Convert the GPP base into higher force production while retaining year-round power exposures, building unilateral and rotational strength, and expanding forearm capacity.

Phase parameters: Five training days across Monday, Tuesday, Thursday, and Saturday, with Wednesday and Friday for recovery. Use heavy loading only on movements with reliable load standards. Complex-training pairings use a full recovery interval between the strength activity and the ballistic activity.

Monday — Heavy Hinge Strength, Vertical Power & Unilateral Force

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Maximal overcoming isometric — lower-body force	Maximal intent against an immovable resistance	3–4 × 3–5 sec	2–3 min	Each effort is brief and maximal; do not turn it into a long hold.
Hinge pattern — maximal strength	82–90% 1RM	3 × 3–4	3–4 min	Use a technically repeatable hinge variation; no grinding.
PAPE contrast: vertical jump pattern	Bodyweight or light external load selected for output	3 × 3	4–7 min after hinge set	Perform only after the full post-conditioning recovery interval; assess whether jump output is enhanced.
Unilateral knee-dominant strength pattern	75–85% of a reliable variation-specific max, or load for 4–6 strong reps	3 × 4–6/leg	2–3 min	Maintain control and consistent range; avoid failure.
Rotational power-generation pattern	Moderate constant-tension load permitting fast concentric action	3 × 4–6/side	90–120 sec	Explosive concentric, controlled return.
Anti-rotation / 360-degree bracing pattern	Moderate load	3 × 6–8 with 2–3-sec holds	60–90 sec	Maintain trunk and pelvic control.

Tuesday — Upper-Body Strength & Forearm Development

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Maximal overcoming isometric — upper-body push	Maximal intent	3 × 3–5 sec	2–3 min	Short, high-intent effort; maintain joint position.
Horizontal push pattern	80–85% 1RM	3–4 × 4–5	2–3 min	Use a stable, repeatable variation; stop short of technical failure.
Horizontal pull pattern	80–85% 1RM where applicable	3–4 × 4–6	2–3 min	Controlled full range; no torso momentum.
Rotational external-rotation strength	Moderate–heavy constant-tension load	3 × 6–8/arm	90 sec	3–4-sec eccentric; strict control.
Wrist flexion capacity	Moderate–heavy standardized load	3 × 6–10	90 sec	Progress load while preserving range and tempo.
Pronation capacity	Moderate standardized load	3 × 8/side	60–90 sec	Controlled rotation through available range.
Supination capacity	Moderate standardized load	3 × 8/side	60–90 sec	Controlled rotation through available range.

Wednesday — Recovery & Positional Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Zone 2 aerobic recovery	Conversational pace	25–35 min	—	Continuous, low-intensity work.
Long-length positional isometric	Submaximal	2 × 30–45 sec	60–90 sec	Use only selected positions; avoid routine 1–2 minute holds.
Mobility / breathing	Low intensity	10–15 min	As needed	Restore movement quality without creating fatigue.

Thursday — Unilateral Strength, Braking & Lateral Power

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Non-axial squat / knee-dominant strength pattern	80–85% of reliable variation-specific max	3–4 × 4	2–3 min	Heavy lower-body force without requiring a barbell back squat.
Eccentric deceleration lunge pattern	Load allowing controlled 3–4-sec lowering	3 × 3/leg	2–3 min	Emphasize lead-leg braking; maintain alignment.
PAPE contrast: lateral bound pattern	Bodyweight or light load	3 × 3/side	4–7 min after eccentric lunge set	Use only if the athlete demonstrates improved or maintained output after the conditioning activity.
Rotational force / anti-rotation pattern	Heavy constant-tension load that preserves position	3 × 5–6/side	90–120 sec	Produce force without losing trunk/pelvis control.
Rotational capsule / cuff strength	Moderate–heavy constant-tension load	3 × 6–8/side	90 sec	4-sec eccentric; strict movement quality.

Friday — Recovery

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Zone 1–2 aerobic flush (optional)	Easy	15–25 min	—	Optional based on recovery and throwing schedule.
Mobility / breathing	Low intensity	10–15 min	As needed	No high-intent work.

Saturday — Upper-Body Strength, Scapular Function & Forearm Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Vertical pull pattern	80–85% 1RM where applicable	3–4 × 4–6	2–3 min	Heavy but technically clean.
Unilateral vertical push pattern	75–80% of reliable variation-specific max	3 × 5/arm	2 min	Use a shoulder-tolerant range and stable trunk.
Scapular upward-rotation pattern	Moderate load	3 × 8–12	60–90 sec	Controlled, full-range repetitions.
Wrist extension capacity	Moderate–heavy standardized load	3 × 8–10	60–90 sec	Balance wrist flexion volume.
Ulnar-deviation capacity	Moderate standardized load	3 × 8–10/side	60–90 sec	Controlled movement.
Finger-flexion capacity	Moderate standardized load	3 × 8–10	60–90 sec	Progress gradually.
Explosive pinch closure	Load permitting fast closure	3 × 5–8	60–90 sec	Maximal speed; terminate on clear speed loss.

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Two- and three-point pinch holds	Moderate, submaximal	2 × 10–15 sec each	60 sec	Maintain clean thumb/finger contact.

3. FIRST TRANSITION / POWER BLOCK (PRE-SEASON)

Primary objective: Convert the strength base into high-rate force production and sport-specific power while preserving structural strength, upper-body speed-strength, scapular capacity, and forearm development.

Phase parameters: Three to four gym / field training exposures. Ballistic loading is selected for velocity and output rather than fatigue. Keep high-output sessions low-volume, allow full recovery, and adjust total work to the athlete's overall training stress.

Monday — Lower-Body RFD, Rotational Power & Acceleration

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Weighted vertical jump pattern	External load approximately 10–20% body mass initially; adjust to preserve jump output	3–4 × 3	2–3 min	Terminate if jump height or output drops >10% from session best.
Rotational medicine-ball power	Light implement selected for maximal speed	3–4 × 3/side	90–120 sec	Maximal intent; reset between repetitions.
Oblique sling / transverse transfer pattern	Light–moderate load	3 × 3–4/side	90 sec	Fast, coordinated movement; no fatigue accumulation.
Unilateral lower-body strength maintenance	80–85% of reliable variation-specific max	2–3 × 3–4/leg	2–3 min	Low volume; preserve force capacity.
Acceleration / sprint pattern	Bodyweight	4–5 × 10 yd	2–3 min	Full recovery; high-quality acceleration.

Tuesday — Upper-Body Speed-Strength, Scapular Force Transfer & Forearm Capacity

This session replaces the former throwing/implement day. Its purpose is to develop rapid upper-body force production while maintaining scapular movement capacity, posterior shoulder strength, and forearm capacity. It is a gym-based training session; no throwing session or throwing progression is prescribed here.

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Explosive horizontal push	Bodyweight plyometric push-up; regress to hands-elevated version if needed	4 × 3	2–3 min	Maximal concentric intent; each rep crisp. Stop if speed or landing quality declines.
Upper-body speed-strength press	Approximately 30–50% 1RM where measurable; select load that preserves speed	3–4 × 3	2–3 min	Fast concentric; controlled return. No grinding or fatigue accumulation.
Speed-strength horizontal pull	Approximately 50–65% of reliable variation-specific max	3 × 4–6	90–120 sec	Accelerate the pull; control the eccentric and maintain trunk position.
Scapular upward rotation / protraction	Light–moderate cable or bodyweight resistance	2 × 8–10	60 sec	Serratus wall slide or cable serratus punch; full, controlled scapular movement.
Scapular retraction / posterior shoulder	Light–moderate resistance	2 × 8–12	60–90 sec	Chest-supported row or prone T raise; avoid excessive fatigue.
Rotator-cuff external-rotation capacity	Light–moderate load	2 × 10–12	45–60 sec	Controlled range; no compensation or painful range.
Forearm pronation / supination and wrist capacity	Light–moderate load	2 × 8–10 per selected action	45–60 sec	Choose 2–3 actions based on current needs; maintain controlled tempo.

Session management: *The explosive push and speed-strength press are the priority. If output declines, remove sets rather than adding repetitions. Keep scapular, cuff, and forearm work submaximal. This session should complement—not duplicate—the Saturday structural-maintenance session.*

Wednesday — Recovery & Positional Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Zone 2 aerobic recovery	Conversational pace	20–30 min	—	Low-intensity aerobic work.
Long-length positional isometric	Submaximal	2 × 30–45 sec	60–90 sec	Short positional dose; avoid long-duration fatigue holds.
Mobility / breathing	Low intensity	10–15 min	As needed	Restore comfortable movement quality.

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Mobility / breathing	Low intensity	10–15 min	As needed	Restore movement quality.

Thursday — KEAT / Shock Method & Explosive Force

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Drop-jump / shock-method pattern	12–18 in starting drop height; progress only when contact quality is maintained	3 × 3–5	2–3 min	Prioritize reactive quality; stop if contact time or jump output deteriorates.
Explosive horizontal / vertical push pattern	Approximately 30–60% 1RM where measurable	3–4 × 3	2–3 min	Maximal concentric intent; terminate on >10% output loss.
Triple-extension ballistic pattern	Light–moderate load selected for speed	3 × 3	2–3 min	Fast execution; no grinding.
Rotational ballistic pattern	Light implement	3 × 3/side	90–120 sec	High intent; full reset.

Friday — Low-Intent Mechanics & Tissue Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Rapid reversal / antagonist timing pattern	Light implement	2–3 × 10–15	60–90 sec	Fast but relaxed; low fatigue.
Forearm flexor-pronator capacity	Light–moderate load	2 × 10–12	60 sec	Wrist flexion, pronation/supination, and finger flexion selected by need.
Posterior cuff / scapular capacity	Light–moderate load	2 × 12–15	45–60 sec	Low fatigue, controlled repetitions.

Saturday — Speed-Strength & Structural Maintenance

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Acceleration pattern	Bodyweight	4 × 10 yd	2–3 min	Full recovery; quality sprint mechanics.
Unilateral lower-body strength maintenance	70–80% of reliable variation-specific max	2 × 5/leg	2 min	Moderate volume; preserve strength.
Horizontal pull pattern	75–80% 1RM where applicable	3 × 6–8	2 min	Maintain upper-back strength.
Forearm capacity circuit	Approximately 65–75% standardized exercise max	2 × 8–10 per selected action	60 sec	Rotate wrist flexion/extension, pronation/supination, deviation, and finger flexion.
Explosive pinch closure	Load permitting fast closure	2 × 5	60 sec	Low-volume neural exposure.

4. COMPETITIVE / IN-SEASON MAINTENANCE

Primary objective: Maintain strength, power, and peripheral tissue capacity with minimum effective volume while prioritizing throwing, games, and recovery.

Phase parameters: Two brief maintenance lifts per week are a starting structure, not a rigid calendar. Move or remove sessions around actual starts, relief appearances, bullpens, and recovery needs.

Maintenance Lift A — Lower-Body Strength & Power

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Vertical jump pattern	Bodyweight or light load	2 × 3	2 min	High quality; stop if output drops.
Hinge pattern — strength maintenance	80–85% 1RM where applicable	2–3 × 3	3 min	Low volume; no grinders.
Unilateral lower-body force maintenance	75–80% of reliable variation-specific max	2 × 4/leg	2 min	Maintain strength with low fatigue.
Rotational ballistic pattern	Light implement	2 × 3/side	90 sec	Fast, crisp repetitions.

Maintenance Lift B — Upper-Body Structure & Forearm Capacity

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Horizontal push pattern	75–80% 1RM where applicable	2–3 × 5	2 min	Moderate volume; leave repetitions in reserve.
Vertical pull pattern	75–80% 1RM where applicable	2–3 × 5	2 min	Controlled, full range.
Wrist flexion capacity	Light–moderate standardized load	2 × 8–10	60 sec	Maintain local capacity.
Wrist extension capacity	Light–moderate standardized load	2 × 8–10	60 sec	Balance flexor volume.
Pinch capacity — 2- and 3-point	Moderate, submaximal	2 × 10–15 sec each	60 sec	Avoid maximal holds immediately before high-intent throwing.

Weekly Power Microdose (Place on a Low-Fatigue Day)

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Vertical jump pattern	Bodyweight or light load	2 × 3	2 min	One brief exposure if throwing schedule permits.
Rotational ballistic pattern	Light implement	2 × 3/side	90 sec	Maintain speed and coordination.
Lateral bound pattern	Bodyweight	2 × 2/side	90–120 sec	Use only when landing quality and recovery are good.

Throwing-Day Tissue Maintenance

Place post-throwing work after the throwing session. The exact dose should reflect the throwing load and athlete response.

Movement / capacity category	Target load / intensity	Sets × reps / duration	Rest	Execution / intent
Flexor-pronator capacity	Light–moderate load	1–2 × 10–12	45–60 sec	Post-throwing local capacity work; adjust to tolerance.
Posterior cuff capacity	Light load	1–2 × 12–15	45–60 sec	Controlled, non-fatiguing.
Elbow flexor / forearm tissue work	Light load	1–2 × 10–15	45–60 sec	Use as a brief maintenance dose, not a separate workout.

5. GLOBAL PRESCRIPTION STANDARDS

Loading and intensity

• Hypertrophy / structural capacity: generally 60–75% 1RM for 6–10 repetitions when a valid 1RM exists.
• General strength: generally 70–80% 1RM for 5–8 repetitions.
• Maximal strength: generally 82–90% 1RM for 3–5 repetitions.
• Heavy strength exposures: generally 85–92% 1RM for 2–4 repetitions when the athlete is prepared and the movement is standardized.
• Ballistic strength: often approximately 30–60% 1RM when a meaningful 1RM exists; otherwise select load by measured velocity, jump height, throw distance, or another repeatable output.
• Do not assign %1RM to non-standardized unilateral, rotational, isometric, or ballistic tasks unless the test is reliable and directly applicable.

Rest-period standards

• Structural / hypertrophy work: 90–150 seconds.
• General strength: 2–3 minutes.
• Heavy strength: 3–5 minutes.
• Ballistic jumps and throws: 90 seconds to 3 minutes, with enough recovery to preserve output.
• Maximal overcoming isometrics: 2–3 minutes between efforts.
• Yielding and positional isometrics: 60–120 seconds depending on load and duration.
• PAPE / complex training: allow approximately 4–7 minutes after the conditioning activity before the ballistic action. The optimal interval is athlete- and task-dependent; test the athlete's response rather than assuming potentiation.

Isometric categories — do not treat all holds the same

• Maximal overcoming isometric: 3–5 sets of 3–5 seconds at maximal intent; rest 2–3 minutes. This is a brief high-force effort against an immovable resistance.
• Yielding isometric: 3–4 sets of 20–30 seconds at moderate-to-heavy effort; rest 60–120 seconds. Maintain the position against a load.
• Long-length positional isometric: 2–3 sets of 30–45 seconds at submaximal effort; rest 60–90 seconds. The purpose is positional tolerance, not maximal force production.
• Avoid defaulting to 60–120-second holds for every isometric category. Duration should match the intended adaptation.

PAPE / complex-training execution

1. Complete the heavy conditioning activity with low total repetition volume.
2. Rest approximately 4–7 minutes before the paired jump, bound, or ballistic action.
3. Measure or observe the paired output where practical (jump height, distance, velocity, contact quality).
4. Keep the pairing only if the athlete maintains or improves the ballistic output. If output is reduced, extend rest, reduce the conditioning load/volume, or separate the activities.
5. Do not force immediate transitions or assume every athlete potentiates at the same interval.

Forearm development — capacity beyond pinch

The weekly forearm plan should expose the athlete to wrist flexion, wrist extension, pronation, supination, ulnar deviation, finger flexion, finger extension, and 2-/3-point pinch. Pinch work remains a specific skill/capacity component, not the entire forearm program. Progress one variable at a time (load, repetitions, hold duration, or speed). Avoid maximal forearm fatigue immediately before high-intent throwing. These exercises develop local capacity; they should not be represented as a guarantee of UCL injury prevention.

Year-round power exposure

Power work is included in every phase: low-volume jumps and rotational ballistics in GPP; heavier strength paired with jumps/bounds in SPP; higher-velocity ballistic and shock-method emphasis in the transition block; and brief low-volume power microdoses in-season. The emphasis and volume change by phase, but power is not absent until the transition block.

Weekly structure at a glance

Phase	Mon	Tue	Wed	Thu	Fri	Sat	Sun
GPP	Lower structural	Upper + forearm	Recovery	Braking + rotation	Recovery	Scapular + forearm	Off
SPP	Heavy lower + power	Upper strength	Recovery	Unilateral + braking	Recovery	Upper + forearm	Off
Transition	RFD + sprint	High-intent throw	Recovery	KEAT / power	Low-intent	Speed-strength	Off / adjust

In-season	Lift A	Throw / game	Recovery	Lift B / adjust	Throw / game	Power microdose / game	Off

Important: The weekly layout is a default template. Actual placement must be coordinated with the athlete's throwing schedule, games, bullpen work, and individual recovery. The strength and conditioning plan should support the throwing plan rather than compete with it.

SALZMAN BASEBALL

GPP & SPP MASTER BLUEPRINT

Version 3 — Spinal/Trunk and Scapular Systems Added

This update preserves the Version 2 weekly training architecture and adds two dedicated exercise-library sections: (1) spinal and trunk mobility, strength, control, and rotational power; and (2) scapular mobility, strength, movement control, and force-transfer capacity.

> ***Evidence and prescription note:*** *Pitching biomechanics studies establish that these regions participate in throwing and are associated with important throwing mechanics. They do not establish that every listed exercise independently increases pitch velocity. Exercise doses below are practical starting prescriptions synthesized from peer-reviewed biomechanics, rehabilitation, EMG, and resistance-training literature—not universally validated pitcher-specific protocols.*

The additions are intended to guide exercise selection. They do not require every pitcher to perform every exercise. Select corrective work from assessment, individual needs, throwing workload, symptoms, and training phase.

6. SPINAL MOBILITY, STABILITY & ROTATIONAL POWER

The spine and trunk should be trained as a coordinated, multi-region system. The cervical, thoracic, lumbar, and sacral/pelvic regions have different structures and movement capacities. The program should develop axial rotation, lateral flexion, flexion/extension control, segmental coordination, and force transmission through the trunk.

> ***Evidence interpretation:*** *Pitching studies commonly quantify trunk or thorax rotation relative to the pelvis; they do not isolate the motion of each individual vertebral segment. Therefore, whole-trunk velocity findings should not be misrepresented as proof that isolated lumbar or cervical motion determines pitch velocity.*

6.1 Why the trunk matters in pitching

Greater maximum trunk rotational velocity has been associated with greater ball velocity and greater elbow varus moment in collegiate pitchers. One study reported that each 100°/s increase above average maximum trunk rotational velocity was associated with a 0.70 m/s increase in ball velocity and a 2.90 N·m increase in elbow varus moment. This supports training rotational performance while monitoring the loads that accompany it. [T1]

Contralateral trunk lean has also been associated with ball velocity and upper-extremity joint moments. This is not a prescription to maximize side bend: more lean may accompany greater velocity while also increasing shoulder and elbow loading. [T2]

Pelvis–trunk sequencing and coordination are important parts of the kinetic chain. The pelvis and thorax rotate in a proximal-to-distal sequence, and studies have linked trunk and pelvic mechanics with ball velocity. [T1, T3]

A systematic review of core training studies found that improvements in core strength/endurance do not consistently transfer to throwing velocity. More dynamic medicine-ball and rotational power approaches appear promising, but the evidence base remains limited and protocols vary. [T4, T5]

6.2 Region-by-region roles and training

Region	Functional role in the system	Training emphasis	Evidence boundary
Cervical spine	Head orientation, visual target acquisition, and coordinated head movement as the trunk rotates and tilts.	Controlled rotation and lateral-flexion range; low-load isometric capacity in multiple directions.	Direct evidence that isolated neck training increases pitch velocity is limited. Treat as head/neck capacity, not a velocity driver. [T6]
Thoracic spine	Upper-trunk orientation and contribution to the rotation measured between the thorax and pelvis; extension and rotation accompany arm positioning.	Active rotation and extension mobility; loaded rotation; rotational medicine-ball power.	Pitching studies support whole-trunk sequencing and rotational velocity, but do not isolate thoracic vertebral contribution. [T1, T3]
Lumbar spine	Transfers and controls forces between thorax and pelvis; participates in flexion, extension, lateral flexion, and limited axial rotation.	Multidirectional strength, anti-extension/anti-rotation control, and tolerance to repeated extension/rotation demands.	Pitching involves substantial lumbar extension; manage cumulative load rather than pursuing maximal lumbar mobility. [T7]
Sacrum / pelvis / SI region	Connects the spine to the pelvis and transfers force between lower extremities and trunk; pelvic rotation and control are relevant to sequencing.	Lumbopelvic coordination, hip-driven pelvic rotation, single-leg pelvic control, and split-stance force transfer.	Do not treat the sacrum as a freely mobile spinal segment or prescribe isolated sacral rotation. Evidence concerns pelvic/trunk mechanics. [T1, T3]

6.3 Lateral flexion — a distinct training quality

Lateral flexion should not be reduced to side-plank endurance. Pitching research has associated contralateral trunk lean with velocity and joint moments. The program should develop the ability to produce, resist, and decelerate side bending, train both sides, and individualize volume where assessment identifies a meaningful deficit. [T2]

Capacity	Exercise category / examples	Starting prescription	Execution
Lateral-flexion isometric endurance	Side plank; side plank with reach	2–3 × 20–30 sec/side	Maintain stacked trunk and pelvis; stop before position deteriorates.
Anti-lateral-flexion strength	Suitcase carry; offset carry	3 × 15–25 m/side	Walk tall without leaning toward or away from the load.
Dynamic lateral-flexion strength	Dumbbell or cable side bend	2–3 × 6–10/side	Controlled 2–3 sec lowering; use a tolerable range.

Capacity	Exercise category / examples	Starting prescription	Execution
Multiplanar power	Diagonal medicine-ball throw; rotational scoop throw	3–4 × 3–5/side	Maximal intent with enough rest to preserve speed.

The relationship between lateral trunk lean and velocity is associative, not proof that deliberately increasing lean improves performance. Do not coach a larger lean as a universal velocity strategy. [T2]

6.4 Spinal and trunk exercise library

Movement category	Exercise options	Starting sets × reps / duration	Purpose
Cervical rotation / lateral flexion	Controlled cervical AROM; gentle isometric rotation; lateral-flexion isometric	1–2 × 5–8 reps; isometrics 2 × 10–15 sec/side	Head/neck movement capacity and low-load strength.
Thoracic rotation	Quadruped thoracic rotation; split-stance cable rotation; half-kneeling landmine rotation	Mobility 2 × 6–8/side; loaded 3 × 6–8/side	Develop active range and strength through rotation.
Thoracic extension	Bench-supported thoracic extension; controlled segmental extension	1–2 × 6–8	Upper-trunk extension movement capacity.
Lumbar flexion / extension control	Cat-camel; bird dog; controlled back extension; rollout progression	1–3 × 6–10	Develop movement control and trunk strength; choose range based on tolerance.
Anti-extension / anti-rotation	Rollout progression; cable anti-rotation press; 360-degree bracing	2–3 × 6–10	Develop force-control capacity; complement, not replace, dynamic rotation.
Lumbopelvic control	Supine pelvic tilt; single-leg stance with pelvic control; contralateral-loaded split squat	2–3 × 6–10 or 20–30 sec	Coordinate pelvis, hip, and trunk under unilateral loading.
Rotational strength	Cable rotation; landmine rotation; diagonal cable lift/chop	3 × 6–8/side	Produce and control force through the transverse plane.
Rotational power	Medicine-ball scoop throw; step-behind throw; shot-put style throw	3–5 × 3–5/side	Develop high-velocity rotational force; stop when speed or coordination falls.

The exercise doses are operational starting points. Mobility and control drills should remain comfortable and controlled; strength exercises should be progressively loaded; ballistic work should be terminated when output or coordination declines. Do not force end-range spinal motion.

6.5 Phase emphasis

Phase	Emphasis	Practical weekly exposure
GPP	Multiplanar movement capacity, trunk hypertrophy/endurance, lateral-flexion strength, pelvic control, introductory rotational power.	2–3 strength exposures; 2 low-volume rotational-power exposures.
SPP	Heavier rotational and lateral-flexion strength; anti-motion capacity; maintain explosive rotation.	2 strength exposures; 1–2 power exposures.
Transition	Rotational speed, diagonal force production, medicine-ball throws, trunk strength maintenance.	2–3 low-volume power exposures; 1 maintenance strength exposure.
In-season	Maintain strength and power with low fatigue; adapt to starts, relief appearances, bullpens, and games.	1–2 brief exposures, adjusted to actual throwing schedule.

6.6 Trunk loading standards

Quality	Starting prescription	Rest
Active mobility	1–2 × 6–10; comfortable controlled range	15–30 sec
Motor control	2–3 × 6–10; light–moderate load	30–60 sec
Isometric capacity	2–3 × 15–30 sec; moderate effort	60–90 sec
Strength / hypertrophy	2–4 × 8–12; challenging, technically controlled	90–150 sec
Maximal strength	3–4 × 4–6; heavy without loss of control	2–3 min
Rotational power	3–5 × 3–5/side; load selected to preserve speed	2–3 min

7. SCAPULAR MOBILITY, STRENGTH & FORCE-TRANSFER SYSTEM

The scapula is a mobile platform that coordinates with the thorax, clavicle, humerus, and rotator cuff. It should not be trained only as a structure to retract, depress, or hold still. The pitcher needs the capacity to protract, retract, upwardly rotate, downwardly rotate, and control anterior/posterior tilt throughout arm elevation and throwing.

The six commonly described scapular motions are upward/downward rotation, internal/external rotation (broadly corresponding to protraction/retraction), and anterior/posterior tilt. These motions occur together rather than as fully independent actions. [S1, S2]

7.1 Scapular function during pitching

Three-dimensional analysis of 25 youth baseball players found that at maximum humeral external rotation the scapula was retracted, upwardly rotated, and posteriorly tilted. At maximum humeral internal rotation it was protracted, remained upwardly rotated, and was anteriorly tilted. This is a key reason not to teach constant retraction or assume downward rotation is required during acceleration. [S1]

In 72 pitchers, the pelvis, thorax, and scapula rotated in a proximal-to-distal sequence. Forward thoracic rotation followed by scapular rotation contributed to the final shoulder external-rotation position; scapular protraction occurred alongside humeral horizontal adduction. [S2]

A study of 33 high-school pitchers found an association between clinically identified scapular dyskinesis and lower shoulder rotation velocity, but it was cross-sectional and does not prove that correcting dyskinesis increases pitch velocity. The study also found no group differences in several other pitching kinematic and kinetic measures. [S3]

A study of youth pitchers reported substantial serratus anterior and upper-trapezius activity during pitching, supporting the role of scapular muscle force couples. Kinetic-chain scapular exercise interventions in baseball players have reported changes in muscle activity, but these should not be described as proof of increased pitch velocity. [S4, S5]

7.2 Four primary movement patterns

Movement	Role in throwing / arm function	Training implication
Protraction	Moves the scapula around the rib cage. Present during acceleration and follow-through; helps coordinate scapular motion with humeral horizontal adduction.	Train serratus anterior and controlled protraction; include concentric and eccentric capacity.
Retraction	Moves the scapula toward the spine. Present around maximum shoulder external rotation/cocking.	Train middle trapezius, rhomboids, and controlled retraction; do not hold the scapula retracted all day.
Upward rotation	Turns the glenoid upward as the arm elevates. Remains present at both measured pitching events in the youth kinematics study.	Train serratus anterior and trapezius force couple; do not treat upward rotation as something to eliminate.
Downward rotation	Returns the scapula toward its resting orientation as the arm lowers; involves coordinated muscular control.	Train controlled return and capacity; do not overemphasize constant depression or forced downward rotation.

7.3 Include anterior and posterior tilt

Posterior tilt and anterior tilt should be included in the scapular model. Posterior tilt commonly accompanies arm elevation and was observed at maximum external rotation in the youth pitching study. Anterior tilt was observed at maximum internal rotation. The serratus anterior and lower trapezius contribute to the force couple involved in upward rotation and posterior tilt. These are coordinated motions, not isolated positions to force. [S1, S6]

7.4 Scapular muscle groups and movement contribution

Muscle / group	Primary contribution	Programming priority
Serratus anterior	Protraction, upward rotation, scapular control against the rib cage.	High
Upper trapezius	Upward rotation and elevation as part of the trapezius force couple.	Moderate–high
Middle trapezius	Retraction and scapular control.	High
Lower trapezius	Upward rotation, posterior tilt, and scapular control.	High
Rhomboids	Retraction and contribution to downward rotation.	Supporting
Levator scapulae	Elevation and contribution to downward rotation.	Supporting
Pectoralis minor	Contributes to anterior tilt, protraction, and downward rotation.	Assess mobility/tension where relevant; do not assume it is always pathological.

A 2024 systematic review of EMG studies in overhead athletes found that scapular exercise research does not consistently emphasize all scapular muscle groups, including the middle trapezius. This supports a more complete program rather than relying almost exclusively on serratus work. EMG findings describe muscle recruitment, not direct performance outcomes. [S7]

7.5 Scapular exercise library

Category	Exercise options	Starting prescription	Purpose
Protraction	Push-up plus; single-arm cable serratus punch; cable dynamic hug.	2–3 × 8–12	Serratus anterior strength and active protraction.
Retraction	Chest-supported row; prone T raise; cable row with controlled eccentric.	2–4 × 8–12	Middle-trapezius/rhomboid capacity and controlled retraction.
Upward rotation	Serratus wall slide; half-kneeling landmine press; wall slide with reach.	2–3 × 8–12 (landmine 3 × 6–10)	Serratus–trapezius force couple during arm elevation.
Downward rotation / controlled return	Scapular pull-up; straight-arm cable pulldown; controlled return from overhead cable reach.	2–3 × 6–12	Controlled return and capacity of muscles contributing to downward rotation.
Posterior tilt / lower trapezius	Prone Y raise; wall slide with lift-off; incline prone elevation with external rotation.	2–3 × 8–12	Lower-trapezius contribution and control during elevation.
Integrated kinetic-chain	Split-stance single-arm cable row; half-kneeling diagonal lift; split-stance landmine press.	2–3 × 6–10/side	Integrate scapular control with trunk and pelvic movement.

These are starting prescriptions, not fixed universal doses. Exercise selection should reflect the capacity being trained, the athlete's symptoms and movement assessment, and the amount of throwing and upper-body strength work already performed. EMG studies support muscle recruitment patterns for several exercises; they do not establish that one exercise is superior for pitch velocity. [S6–S9]

7.6 Phase emphasis and weekly integration

Phase	Emphasis	Practical prescription
GPP	Movement capacity, protraction/retraction strength, upward rotation, muscular endurance.	2–3 exercises/session; generally 2–3 × 10–15.
SPP	Heavier scapular strength, eccentric control, and force production.	2–3 exercises/session; generally 3–4 × 6–10.
Transition	Maintain strength while integrating scapular movement into throwing and upper-body power.	1–2 strength exercises; 2–3 × 6–10; ballistic work low volume.
In-season	Maintain strength, movement quality, and endurance with minimum effective volume.	1–2 sessions/week; 1–3 exercises/session; generally 2 × 8–12.

Suggested integration with the existing Version 2 week

Day	Suggested scapular work	Programming note
Tuesday — upper-body day	Push-up plus 2 × 10–12; chest-supported row 3 × 8–10; prone Y raise 2 × 10; serratus wall slide 2 × 10.	Use as selected components of the existing upper-body session, not automatically as additional volume on top of all existing work.
Saturday — upper-body day	Single-arm cable row 3 × 8/side; half-kneeling landmine press 3 × 6–8/side; cable serratus punch 2 × 10/side; controlled scapular pull-up 2 × 8.	Adjust volume around throwing intensity and fatigue; avoid unnecessary fatigue immediately before high-intent throwing.

7.7 Warm-up, corrective, and recovery applications

Use	Exercise	Dose	Selection rule
Warm-up	Serratus wall slide; scapular push-up; light cable row.	1–2 × 8–10	Use movements that improve readiness without fatigue.
Corrective	Prone Y raise; push-up plus; controlled cable row.	2–3 × 8–12	Assessment-driven; choose for a specific capacity deficit.
Recovery / movement	Gentle wall slide; scapular circles.	1–2 × 6–8	Comfortable active movement; not a mandatory recovery ritual.

7.8 Scapular programming principles

• Do not permanently hold the scapula retracted; throwing requires both retraction and protraction. [S1, S2]
• Do not treat upward rotation as something to eliminate; it is present at important points in the pitching cycle. [S1]
• Do not equate scapular depression with stability; the scapula must upwardly rotate during arm elevation. [S6]
• Train concentric and eccentric control, not only static holds. [S6, S8]
• Integrate scapular work with the trunk, pelvis, and upper extremity. [S2, S4, S5]
• Do not promise velocity gains from corrective exercises; evidence supports biomechanical relevance and muscle capacity, but direct velocity-intervention evidence remains limited. [S3, S5]

> *Assessment caveat: visible scapular asymmetry or a clinical dyskinesis label alone does not establish a defect requiring correction. Use symptoms, function, throwing demands, strength, and movement assessment to guide exercise selection. [S3]*

8. PEER-REVIEWED REFERENCE LIST — NEW SECTIONS

Spine and trunk

[T1] Aguinaldo AL, et al. The Association Among Trunk Rotation, Ball Velocity, and the Elbow Varus Moment in Collegiate-Level Baseball Pitchers. 2019. PubMed: https://pubmed.ncbi.nlm.nih.gov/31424975/

[T2] Lateral Trunk Lean in Pitchers Affects Both Ball Velocity and Upper Extremity Joint Moments. 2015. 99-pitcher descriptive laboratory study; per 10° greater contralateral lean, ball velocity was 0.5 m/s higher, elbow varus moment 3.7 N·m higher, and glenohumeral internal-rotation moment 2.5 N·m higher. DOI: 10.1177/0363546515574060. PubMed: https://pubmed.ncbi.nlm.nih.gov/25762150/

[T3] Orishimo KF, Kremenic IJ, Mullaney MJ, Fukunaga T, Serio N, McHugh MP. Role of Pelvis and Trunk Biomechanics in Generating Ball Velocity in Baseball Pitching. *Journal of Strength and Conditioning Research*. 2023;37(3):623–628. DOI: 10.1519/JSC.0000000000004314. PubMed: https://pubmed.ncbi.nlm.nih.gov/35836313/

[T4] Oyama S, Palmer TG. Effectiveness of Core Exercise Training Programs Designed to Enhance Ball-Throwing Velocity in Overhead Athletes: A Systematic Review. *Strength & Conditioning Journal*. 2023;45(2):177–187. DOI: 10.1519/SSC.0000000000000738. https://doi.org/10.1519/SSC.0000000000000738

[T5] Enhancing Overhead Throwing Ball Velocity After Core Muscle Training in Athletes: A Systematic Review and Meta-analysis of Randomized Controlled Trials. 2024. DOI: 10.1097/PHM.0000000000002469. PubMed: https://pubmed.ncbi.nlm.nih.gov/38529609/

[T6] Sagittal Plane Trunk Tilt Is Associated With Upper Extremity Joint Moments and Ball Velocity in Collegiate Baseball Pitchers. 2018. 99 collegiate pitchers; trunk tilt at release was associated with ball velocity and elbow varus moment. PubMed: https://pubmed.ncbi.nlm.nih.gov/30349837/

[T7] Lumbar hyperextension in baseball pitching. 2018. PubMed: https://pubmed.ncbi.nlm.nih.gov/29809066/

Evidence note: T1–T3 and T6 are pitching biomechanics/association studies. T4–T5 synthesize intervention studies in overhead throwers. These sources support the relevance of trunk function and some training approaches; they do not validate every exercise or dose in this blueprint.

8. PEER-REVIEWED REFERENCE LIST — CONTINUED

Scapular movement and training

[S1] Oliver G, Weimar W. Scapula Kinematics of Youth Baseball Players. *Journal of Human Kinetics*. 2015;49:47–54. DOI: 10.1515/hukin-2015-0107. Full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC4723181/

[S2] Yanai T, et al. Proximal to distal sequencing impacts on maximum shoulder joint angles and the risk of impingement in baseball pitching involving a scapular independent thoracohumeral model. *Scandinavian Journal of Medicine & Science in Sports*. 2023. DOI: 10.1111/sms.14374. PubMed: https://pubmed.ncbi.nlm.nih.gov/37073523/

[S3] Relationship Between Clinical Scapular Assessment and Scapula Resting Position, Shoulder Strength, and Baseball Pitching Kinematics and Kinetics. *Orthopaedic Journal of Sports Medicine*. 2021. DOI: 10.1177/2325967121991146. Full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC8239341/

[S4] Gluteus medius and scapula muscle activations in youth baseball pitchers. 2015. PubMed: https://pubmed.ncbi.nlm.nih.gov/25546449/

[S5] Song KJ, Yoon JH, Oh JK. Effects of Scapular Kinetic-Chain Exercise on Muscle Activity in Overhead-Pitching Baseball Players. *Iranian Journal of Public Health*. 2020;49(5):875–885. PMID: 32953675. Full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC7475635/

[S6] Kibler WB, Sciascia A. Evaluation and Management of Scapular Dyskinesis in Overhead Athletes. Clinical review. 2019. Use as clinical background for scapular assessment and management; not direct proof of velocity effects.

Evidence note: S1–S2 describe scapular kinematics and sequencing during pitching. S3 is a cross-sectional clinical/biomechanical study. S4 describes muscle activation during pitching. S5 is an 8-week exercise intervention with muscle-activation outcomes. None establishes that a particular scapular exercise independently increases pitch velocity.

Prescription standard

The sets, repetitions, durations, and phase recommendations in Sections 6 and 7 are practical starting prescriptions. Where a study reports only biomechanics, EMG, or association, the blueprint identifies that limitation rather than treating it as direct intervention evidence. Individualize exercise choice and volume to the athlete's assessment, symptoms, throwing workload, and recovery.